Exploring ML Models for Detection of Atmospheric Composition in Exoplanets

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A. Objective And Significance

The goal of this project is to build and evaluate machine learning models that can rapidly predict the atmospheric chemical composition of rocky exoplanets from simulated climate profiles. Specifically, we predict the $\log_{10}$ surface volume mixing ratios of twelve molecular species — including key biosignatures oxygen (O_2), ozone (O_3), methane (CH_4), and nitrous oxide (N_2O) — from a (12×101) dimensional CLIMA atmospheric profile tensor. This is a multi-output regression task with twelve simultaneous continuous targets.

The scientific motivation is urgent. The James Webb Space Telescope (JWST), launched in December 2021, is actively observing exoplanet transmission spectra at unprecedented sensitivity. The planned Habitable Worlds Observatory (HWO) will survey thousands of rocky planets. Traditional retrieval methods based on Bayesian sampling (MCMC or nested sampling) require thousands to hundreds of thousands of likelihood evaluations per planet, often taking hours of compute time, making population-level analysis infeasible. A trained ML model, by contrast, infers all twelve abundances in milliseconds, enabling the fast automated screening that next-generation surveys demand.

Related work has demonstrated the viability of ML for retrieval. Márquez-Neila et al. (2018) applied random forests to brown dwarf spectra. Vasist et al. (2023) used normalising flows for posterior estimation. Gebhard et al. (2024) developed flow matching for full posterior retrieval. Most relevantly, Zorzan et al. (2025) released the INARA dataset — the first large-scale ML benchmark for rocky planet atmospheric retrieval, containing 3.1 million synthetic CLIMA profiles. Our work applies a compact 1D CNN directly to the INARA ATMOS profile representation, treating altitude structure as a sequence rather than a flat feature vector, and compares it systematically against a PCA + Random Forest baseline on 124,320 samples executed on Northeastern Explorer HPC.

B. Background

B.1. Atmospheric Retrieval and Transit Spectroscopy

When an exoplanet transits in front of its host star, starlight filtered through the atmosphere carries molecular absorption fingerprints. The transmission spectrum encodes atmospheric chemical composition. Recovering molecular abundances from this spectrum is called atmospheric retrieval. The benchmark observation is WASP-39 b (JWST ERS Team, 2023), which confirmed CO_2, SO_2, H_2O, and CO in a single observation, demonstrating JWST's capability across the key biosignature wavelength range. Classical retrieval uses MCMC or nested sampling to sample the posterior distribution. While rigorous, these methods require hours of compute per planet, making population-level surveys infeasible.

B.2. Previous Machine Learning Work

Márquez-Neila et al. (2018) showed that random forests trained on synthetic spectra can recover atmospheric parameters with R^2 comparable to Bayesian methods at a fraction of the compute cost. Vasist et al. (2023) used normalising flows for full posterior estimation. Gebhard et al. (2024) demonstrated flow matching for multimodal posteriors. Our approach targets point-estimate retrieval at maximum throughput, appropriate for rapid screening of large planetary populations.

B.3. INARA Dataset and What Makes This Work Interesting

Zorzan et al. (2025) released the INARA dataset — 3.1 million synthetic rocky planet simulations from the CLIMA photochemical-climate model, spanning F, G, K, and M stellar types. Each simulation produces a (12×101) CLIMA profile: twelve atmospheric variables at 101 altitude levels. This altitude-resolved profile captures vertical structure invisible in wavelength-domain spectra. Our contribution treats this altitude dimension explicitly as a sequence: a 1D CNN with stride-2 convolutions progressively compresses the 101-level altitude axis while learning hierarchical patterns. This is qualitatively different from the PCA+RF baseline, which discards all spatial ordering by flattening and rotating the input.

C. Methods

C.1. Data and Preprocessing

Dataset. We use the INARA ATMOS subset (Zorzan et al. 2025). Raw data consists of 11 tar.gz archives (dir-0 through dir-9 plus dir-alpha), ~60 GB total. Each planet is stored as four files: (1) parsed-clima-final.npy.npz (the 12×101 CLIMA profile); (2) parsed-photochem-mixing-ratios.npy.npz; (3) mixing-ratios.dat (N_2 surface VMR); and (4) run-metadata.json. Only planets with all four files present are included (~39% of hashes in any archive are complete).

Input features. Each (12×101) CLIMA tensor is used directly after per-channel Z-score normalisation (statistics fit on training set only). The 12 channels are: actinic flux, pressure, altitude, temperature, convective flux, temperature tendency, previous temperature, water vapour flux, saved radiative flux, ozone flux, radiative cooling, and radiative heating.

Target extraction. The surface row (altitude index 0) of the photochemical mixing ratio array provides VMR for each of 12 target molecules. All values are converted to $\log_{10}$ with a detection floor of −40.0. Targets are Z-score normalised per molecule (MoleculeScaler) during training and inverse-transformed for evaluation.

Data split. 124,320 samples split 70/15/15 (train/val/test), fixed seed 42: 86,924 train / 18,634 val / 18,762 test.

Feature engineering rationale. Minimal feature engineering is required because CLIMA profiles are physically structured — each channel has a distinct physical meaning and all 101 altitude levels are informative. Z-score normalisation and $\log_{10}$ target compression are the only transformations. The RF baseline additionally applies PCA(300 components) after flattening, retaining ≥95% of variance.

C.2. Baseline Model — Per-Molecule Random Forest

Twelve independent RandomForestRegressor models, one per target molecule. Pipeline: flatten (12×101)→(1,212,), apply PCA(300 components), feed to each regressor. Hyperparameters: n_estimators=100, max_depth=8, min_samples_leaf=2, max_features='sqrt'. Training capped at 10,000 samples per molecule (memory constraint: full PCA matrix at 86K samples exceeds available RAM). Inference stacks 12 independent predictions into a (12,) output vector.

C.3. Deep Model — Compact 1D CNN

The CNN operates directly on the (B, 12, 101) tensor, treating 101 altitude levels as a sequence. Architecture:

Layer	Output Shape	Components
Input	(B, 12, 101)	Z-score normalised CLIMA profile
Conv Block 1	(B, 32, 25)	Conv1d(12→32, k=9, s=2, p=4) + BN + ReLU + MaxPool1d(2)
Conv Block 2	(B, 64, 6)	Conv1d(32→64, k=7, s=2, p=3) + BN + ReLU + MaxPool1d(2)
Conv Block 3	(B, 128, 1)	Conv1d(64→128, k=5, s=2, p=2) + BN + ReLU + MaxPool1d(2)
Conv Block 4	(B, 256, 1)	Conv1d(128→256, k=3, s=1, p=1) + BN + ReLU [no MaxPool]
GlobalAvgPool	(B, 256)	AdaptiveAvgPool1d(1) + Flatten
Shared FC	(B, 128)	Dropout(0.25) + Linear(256→128) + LayerNorm + ReLU
12 × Heads	(B, 12)	Per-molecule MLP: FC(128→hidden→1) + LN + ReLU + Dropout → $\log_{10}$ VMR

Table 1. CNN1D architecture. BN = BatchNorm1d. Total trainable parameters: 368,588.

Stride-2 convolutions and MaxPool progressively halve the sequence (101→25→6→1). AdaptiveAvgPool1d(1) eliminates the sequence dimension producing a 256-dim global feature vector. Per-molecule heads prevent gradient interference between molecules with very different target variances (e.g. NH_3 at constant −40 vs H_2 spanning 10 log-orders).

C.4. Training Configuration

Hyperparameter	Value
Optimizer	AdamW (lr=1e−3, weight-decay=1e−4)
LR Scheduler	CosineAnnealingLR (T-max=150, η-min=1e−6)
Max epochs	150
Early stop patience	30 epochs (min-delta=1e−5, monitors val-loss)
Batch size	32 (train) / 64 (eval)
Gradient clipping	clip-grad-norm- max-norm=5.0
Noise augmentation	Gaussian noise std=0.01 on training inputs only
Loss function	WeightedMSE with per-molecule importance weights
Target scaling	Z-score per molecule (MoleculeScaler)

Table 2. CNN1D training configuration (HPC run).

AdamW applies weight decay directly to weights (decoupled from adaptive gradient), giving better generalisation than standard Adam. **CosineAnnealingLR** smoothly decays LR from 1e−3 to 1e−6 following a cosine curve — fast early descent, smooth fine-tuning near convergence. **Gradient clipping** (max-norm=5.0) prevents exploding gradient spikes from simultaneous multi-head backpropagation through the shared backbone. **Gaussian noise augmentation** (std=0.01, train only) prevents memorisation of exact profile values and acts as implicit input regularisation.

C.5. Evaluation Strategy

Both models evaluated on the identical held-out test set (18,762 samples). Three metrics per molecule: (1) **R^2** — standard retrieval quality metric, directly comparable to Bayesian benchmarks; (2) **RMSE** in $\log_{10}$ VMR units (1.0 = one order-of-magnitude error); (3) **MAE** — robust to log-floor outliers. Per-molecule metrics reported for all 12 targets plus a MEAN summary row. Caveats: NH_3 is at log-floor (−40) for all samples — R^2=1.0 is trivial for both models. H_2O has near-zero variance; R^2 is unreliable for that molecule specifically.

D. Results

D.1. Experiment 1 — Baseline vs CNN1D on HPC 124K Dataset

Both models evaluated on 18,762 held-out test samples. RF trained on 10,000 samples; CNN1D on 86,924. Table 3 reports per-molecule R^2, RMSE, and MAE.

Molecule	RF R^2	RF RMSE	CNN R^2	CNN RMSE	CNN MAE	ΔR^2
H_2O	0.614	0.000024	0.979	0.000024	0.0000017	+0.365
CO_2	0.888	0.282	0.9998	0.012	0.008	+0.112
O_2	0.728	0.090	0.9987	0.006	0.004	+0.271
O_3	0.752	0.062	0.9990	0.004	0.002	+0.247
CH_4	0.928	0.493	0.9999	0.009	0.006	+0.072
N_2	0.709	0.023	0.9979	0.002	0.001	+0.289
N_2O	0.762	0.075	0.9992	0.004	0.003	+0.237
CO	0.934	0.345	0.9999	0.011	0.009	+0.066
H_2	0.751	1.268	0.9984	0.103	0.021	+0.247
H_2S	0.945	0.093	0.9999	0.003	0.002	+0.055
SO_2	0.899	0.035	0.9998	0.002	0.001	+0.101
NH_3	1.000	0.000	1.000	0.000	0.000	0.000
MEAN	**0.826**	**0.231**	**0.998**	**0.013**	**0.005**	**+0.172**

Table 3. Per-molecule test results, HPC 124K run. RMSE and MAE in $\log_{10}$ VMR units.

CNN1D achieves near-perfect retrieval ($R^2 \geq 0.997$) for 11 of 12 molecules, mean R^2=0.998. RF achieves mean R^2=0.826. All four biosignature molecules — O_2 (0.999), O_3 (0.999), CH_4 (0.9999), N_2O (0.999) — are retrieved near-perfectly by the CNN1D.

D.2. Experiment 2 — Validation vs Test: No Overfitting Evidence

Table 4 compares validation and test R^2 per molecule. A negligible gap confirms the model generalises and did not overfit to the validation set.

Molecule	Val R^2	Test R^2	|Val−Test|
H_2O	0.9991	0.9788	0.0203
CO_2	0.9998	0.9998	0.0000

O_2	0.9987	0.9987	0.0000
O_3	0.9989	0.9990	0.0001
CH_4	0.9997	0.9999	0.0002
N_2	0.9982	0.9979	0.0003
N_2O	0.9992	0.9992	0.0000
CO	0.9996	0.9999	0.0003
H_2	0.9986	0.9984	0.0002
H_2S	0.9998	0.9999	0.0001
SO_2	0.9998	0.9998	0.0000
NH_3	1.0000	1.0000	0.0000
MEAN	**0.9993**	**0.9976**	**0.0017**

Table 4. Val vs test R^2. Mean gap 0.0017 — negligible, confirms no overfitting.

The only notable gap is H_2O (0.0203), explained by near-zero target variance, not by overfitting. The val/train loss ratio converges toward 1.0 over 150 epochs with no upward divergence at any point.

D.3. Experiment 3 — Training Dynamics

Train loss converges from 0.091 at epoch 1 to 0.008 at epoch 150. Val loss converges from 0.027 to 0.0009, with best checkpoint at epoch 138 (val-loss=0.000899). Two phases are visible on a log scale: rapid descent in epochs 1–40 (LR near 1e−3) and slow fine-tuning in epochs 40–150 (LR decaying toward 1e−6). No divergence between train and val curves observed at any point.

D.4. Analysis — Anomalous Molecules

H[SUB:2O.] RF achieves R^2=0.614 despite RMSE≈0.0001. This arises because H_2O $\log_{10}$ target has near-zero variance (σ≈0.0002), making the R^2 denominator (SS_{tot}) near zero. The CNN achieves R^2=0.979 by capturing the tiny residual variation the RF misses. RMSE is the honest metric for H_2O.

NH[SUB:3.] Both models achieve R^2=1.000. This is degenerate: NH_3 VMR is at the log-floor (−40) for all samples. Both models learn to predict −40 always, trivially achieving zero residual. Retained for schema compatibility only.

D.5. Computational Efficiency

CNN1D processes 64 planets in ~2–4 ms on GPU (≥15,000 planets/second). RF requires ~10 ms per planet (sequential across 12 models). Classical Bayesian retrieval requires hours per planet. CNN1D achieves ~10^6× speedup over Bayesian methods while exceeding RF by +21% mean R^2.

E. Conclusions

We demonstrated that a compact 1D CNN with a shared convolutional backbone and per-molecule output heads recovers atmospheric chemical abundances near-perfectly (mean R^2=0.998, 11/12 molecules at $R^2 \geq 0.997$) on 124,320 INARA ATMOS samples. The model outperforms a PCA + Random Forest baseline (mean R^2=0.826) by +21%, and retrieves all four biosignature molecules (O_2, O_3, CH_4, N_2O) near-perfectly. Val/test R^2 gap of 0.0017 confirms no overfitting.

Key technical lessons: treating altitude as a sequence (CNN) rather than flattening (RF) substantially improves retrieval. The regularisation stack (AdamW, Dropout, BatchNorm, noise augmentation, gradient clipping) was essential for stable training across 12 simultaneous regression heads with very different target variances.

What did not work as intended: (1) NH_3 cannot be learned from CLIMA on this dataset version — degenerate target. (2) RF comparison is not data-equal (10K vs 86K), so the gap conflates architecture and data-scale effects. (3) All experiments are on synthetic data; domain transfer to real JWST spectra is unvalidated.

Concrete future directions:

- Train on the full 3.1M INARA dataset on Northeastern Explorer HPC for maximum coverage.
- Add Monte Carlo Dropout or deep ensembles for calibrated uncertainty quantification.
- Validate on real JWST transmission spectra (e.g. WASP-39 b) to confirm domain transfer.
- Run a data-equal ablation (RF at 86K samples) to isolate the pure architecture benefit.

F. Individual Tasks and Team Contributions

Shantanu Wankhare contributed on the topic research and project scoping phase. Shantanu identified the INARA dataset and the exoplanet atmospheric retrieval problem as the project focus, reviewed the relevant literature (Zorzan et al. 2025, Márquez-Neila et al. 2018, Vasist et al. 2023), and established the scientific framing, connecting the ML task to JWST observations and biosignature detection. Shantanu contributed to the initial EDA notebook, helped define the evaluation metrics, and participated in weekly design discussions. Further he contributed in ML model code sub-parts and execution.

Bhalchandra Shinde contributed to the overall design and implementation of the project work. Bhalchandra implemented the deep learning model CNN1D architecture. Bhalchandra designed the pipeline and explored the HPC execution. He ran both the local development runs (10k, 20k, 25K, 124k samples on MacBook M-series CPU) and the full HPC production runs (20k, 124K samples on Northeastern Discovery GPU cluster). Bhalchandra also built the pipeline for local as well as the HPC run, produced all final visualisations (CNN architecture diagram, RF decision tree, training history charts, result comparison figures), and assembled the project documentation, presentation, and this report. Explored the dashboard options and designed, implemented the dashboard.

Asad Mulani contributed to data acquisition and the baseline model. Asad handled obtaining the INARA source data archives and understanding their structure (the 11 tar.gz archives, per-planet file layout, and extraction requirements). Asad implemented the per-molecule Random Forest baseline (baseline-model.py), including the PCA preprocessing pipeline and the 12-model training loop. Asad also supported the HPC job execution, assisting with SLURM job submission and monitoring for the production 124K-sample runs, and contributed to the feature importance analysis for the RF model.

All three team members participated in defining the problem formulation, discussing results interpretation (H_2O variance artefact, NH_3 degeneracy), conducted all hyperparameter selection (architecture depth, kernel sizes, dropout rate, learning rate) and preparing the final presentation and report. Workflow was managed through a shared Git repository with feature branches; design decisions were made jointly in weekly team meetings.

G. REFERENCES

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